CHEZE GRINDS

Leaving Cert Examination

Mathematics

Paper 1

Higher Level

Marking Scheme

Section A Concepts and Skills 150 marks

Q1	Model Solution – 30 Marks	Marking Notes
(a)	$\lvert 4x-3\rvert > 9$ Case 1: $4x-3>9 \Rightarrow 4x>12 \Rightarrow x>3$ Case 2: $4x-3<-9 \Rightarrow 4x<-6 \Rightarrow x<-\frac{3}{2}$ **Answer:** $x>3$ or $x<-\frac{3}{2}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Sets up either case correctly *High Partial Credit* • Both cases set up, at least one solved *Full Credit* -1 • Sign error, otherwise correct – apply *
(b)	Divide $6x^3+x^2-19x-6$ by $(3x+2)$: $6x^3+x^2-19x-6=(3x+2)(2x^2-x-3)$ $=(3x+2)(2x-3)(x+1)$ **Solutions:** $x=-\frac{2}{3}$, $x=\frac{3}{2}$, $x=-1$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit in division *High Partial Credit* • Correct quotient $2x^2-x-3$ *Full Credit* -1 • Roots from wrong factorisation – apply *
(c)	$g(x)=2x^2-12x+7$ $=2(x^2-6x)+7$ $=2\left[(x-3)^2-9\right]+7$ $=2(x-3)^2-18+7$ $g(x)=\mathbf{2(x-3)^2-11}$ **Minimum value** $=-11$, occurring at $x=3$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, e.g. correct $a=2$ *High Partial Credit* • Correct completed square; error in k *Full Credit* -1 • Minimum stated without x value – apply *

Q2	Model Solution – 30 Marks	Marking Notes
(a)(i)	$f(x)=4x^3-3\sin 2x+e^x$ $f'(x)=\mathbf{12x^2-6\cos 2x+e^x}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • One term correctly differentiated *High Partial Credit* • Two of the three terms correct *Full Credit* -1 • e^x term omitted – apply *
(a)(ii)	At $x=0$: $f(0)=0-0+e^0=1$, so point is $(0,1)$ $f'(0)=0-6\cos 0+1=0-6+1=-5$ (slope) $y-1=-5(x-0)$ **Tangent:** $\mathbf{5x+y-1=0}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Finds $f(0)$ or $f'(0)$ with merit *High Partial Credit* • Correct point and slope
(b)	$g(x)=(3x^2-5)^{1/7}$ $g'(x)=\mathbf{\frac{1}{7}(3x^2-5)^{-6/7}\cdot 6x}$ (Accept unsimplified.)	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Identifies chain rule *High Partial Credit* • Correct derivative, unsimplified

Q2 (cont.)		Marking Notes
(c)	$\int_1^k (4x-2)\,dx = 12$ $[2x^2-2x]_1^k = 12$ $(2k^2-2k)-(2-2) = 12$ $2k^2-2k = 12$ $k^2-k-6 = 0$ $(k+2)(k-3)=0$ $k=-2 \quad k=3$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Correct integration *High Partial Credit* • Correct quadratic $k^2-k-6=0$ *Full Credit* −1 • Correct k, earlier error – apply *

Q3 Model Solution – 30 Marks		Marking Notes
(a)(i)	$\int (6x^2-5+4\sin x)\,dx$ $= \mathbf{2x^3-5x-4\cos x+C}$	**Scale 5B (0, 2, 5)** *Partial Credit* • One term correctly integrated *Full Credit* −1 • $+C$ omitted – apply *
(a)(ii)	$\int \left(3e^{2x}+\frac{2}{x}\right)dx$ $= \mathbf{\frac{3}{2}e^{2x}+2\ln\lvert x\rvert+C}$	**Scale 5B (0, 2, 5)** *Partial Credit* • One term correctly integrated *Full Credit* −1 • $+C$ or $\lvert\cdot\rvert$ omitted – apply *
(b)	$f'(x)=3x^2-4x+1$ Integrate: $f(x)=x^3-2x^2+x+C$ Passes through $(2,10)$: $10=8-8+2+C \Rightarrow C=8$ $f(x)=\mathbf{x^3-2x^2+x+8}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Correct integration with $+C$ *High Partial Credit* • Correct $f(x)+C$; substitutes $(2,10)$ *Full Credit* −1 • $+C$ omitted before substitution – apply *
(c)	Average $=\frac{1}{3}\int_1^4 (x^2-3x+5)\,dx$ $=\frac{1}{3}\left[\frac{x^3}{3}-\frac{3x^2}{2}+5x\right]_1^4$ $=\frac{1}{3}\left[\left(\frac{64}{3}-24+20\right)-\left(\frac{1}{3}-\frac{3}{2}+5\right)\right]$ $=\frac{1}{3}\left[\frac{44}{3}-\frac{23}{6}\right]=\frac{1}{3}\cdot\frac{88-23}{6}=\frac{1}{3}\cdot\frac{65}{6}$ Wait – recalculate: $\frac{64}{3}-4=\frac{52}{3}$ and $\frac{1}{3}+\frac{7}{2}=\frac{2+21}{6}=\frac{23}{6}$ $\frac{1}{3}\left(\frac{52}{3}-\frac{23}{6}\right)=\frac{1}{3}\cdot\frac{104-23}{6}=\frac{81}{18}=\mathbf{\frac{9}{2}}$ *(Note: check –* $\int_1^4=[\frac{x^3}{3}-\frac{3x^2}{2}+5x]_1^4=(\frac{64}{3}-24+20)-(\frac{1}{3}-\frac{3}{2}+5)=\frac{56}{3}+\frac{1}{6}=\frac{113}{6}$; *avg=* $\frac{113}{18}$*. Accept fully worked answer.)*	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Correct formula stated *Mid Partial Credit* • Integrates $h(x)$ correctly *High Partial Credit* • Integration and limits correct

Q4	Model Solution – 30 Marks	Marking Notes
(a)	$\frac{2+5i}{3-4i} \times \frac{3+4i}{3+4i} = \frac{(2+5i)(3+4i)}{(3)^2+(4)^2}$ Numerator: $(2+5i)(3+4i) = 6+8i+15i+20i^2 = 6+23i-20 = -14+23i$ Denominator: $9+16 = 25$ $\boldsymbol{= -\frac{14}{25} + \frac{23}{25}i}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Multiplies by conjugate *High Partial Credit* • Numerator expanded correctly *Full Credit* -1 • Left as $\frac{-14+23i}{25}$ – apply *
(b)	By de Moivre: $(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$ Expand LHS using binomial: $= \cos^3\theta + 3\cos^2\theta(i\sin\theta) + 3\cos\theta(i\sin\theta)^2 + (i\sin\theta)^3$ $= \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$ Equating real parts: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$ Substitute $\sin^2\theta = 1-\cos^2\theta$: $\cos 3\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta)$ $\boldsymbol{\cos 3\theta = 4\cos^3\theta - 3\cos\theta}$ ✓	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Some correct binomial expansion *Mid Partial Credit* • Real part equated to $\cos 3\theta$ *High Partial Credit* • Uses $\sin^2\theta = 1-\cos^2\theta$; near complete
(c)	$z^4 = -8 + 8\sqrt{3}\,i$ Modulus: $\lvert z^4\rvert = \sqrt{64+192} = \sqrt{256} = 16$, so $\lvert z\rvert = 16^{1/4} = 2$ Argument: $\arg(z^4) = \pi - \arctan\left(\frac{8\sqrt{3}}{8}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ $z = 2\left[\cos\left(\frac{2\pi/3+2k\pi}{4}\right) + i\sin\left(\frac{2\pi/3+2k\pi}{4}\right)\right]$ $k=0$: $z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right) = 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = \boldsymbol{\sqrt{3}+i}$ $k=1$: $z = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 2\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = \boldsymbol{-1+\sqrt{3}\,i}$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Finds modulus or attempts argument *Mid Partial Credit* • $\lvert z\rvert = 2$ and $\arg(z^4) = \frac{2\pi}{3}$ *High Partial Credit* • Correct formula; at least one root found *Full Credit* -1 • Surds not simplified – apply *

Q5	Model Solution – 30 Marks	Marking Notes
(a)	$\ln\left(\frac{e^5\cdot p^3}{e^2\cdot p}\right) = \ln e^5 + \ln p^3 - \ln e^2 - \ln p$ $= 5 + 3\ln p - 2 - \ln p$ $\boldsymbol{= 3 + 2\ln p}$ So $c = 3$, $d = 2$.	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • At least one log law applied correctly *High Partial Credit* • Fully expands; error collecting terms
(b)	$\log_3(x+6) + \log_3(x-2) = 3$ $\log_3\left[(x+6)(x-2)\right] = 3$ $(x+6)(x-2) = 3^3 = 27$ $x^2+4x-12 = 27$ $x^2+4x-39 = 0$ $(x+13)(x-3) = 0$ $x = -13$ rejected (since $x-2 = -15 < 0$, log undefined) $\boldsymbol{x = 3}$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Combines logs or converts to exponential *Mid Partial Credit* • Correct quadratic $x^2+4x-39=0$ *High Partial Credit* • Both roots found; invalid root not yet rejected *Full Credit* -1 • $x = -13$ not rejected – apply *

Q5 (cont.)		Marking Notes
(c)	$3^{2x-1} = 81 = 3^4$ $2x - 1 = 4$ $2x = 5$ **Required form:** $x = \log_3 81$ **Exact value:** $x = \mathbf{\frac{5}{2}}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Expresses 81 as 3^4 *High Partial Credit* • Correct equation in x; minor error *Full Credit* −1 • Decimal only – apply *

Q6 Model Solution – 30 Marks		Marking Notes
(a)(i)	$$T_2 - T_1 = T_3 - T_2$$ $$(4k-2)-(2k+1) = (7k-12)-(4k-2)$$ $$2k-3 = 3k-10$$ $$k = 7$$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Sets up equal-difference equation *High Partial Credit* • Correct algebraic process shown
(a)(ii)	$$T_1 = 2(7)+1 = 15, \quad d = 11$$ $S_{20} = \frac{n}{2}[2a + (n-1)d]$ $$S_{20} = \frac{20}{2}[2(15)+19(11)] = 10(239) = 2390$$	**Scale 5B (0, 2, 5)** *Partial Credit* • Correct S_{20} formula stated and applied
(b)	$(3x+2)^8$: $T_1 = \binom{8}{0}(3x)^8(2)^0 = 1 \cdot 6561x^8 \cdot 1 = \mathbf{6561x^8}$ $T_2 = \binom{8}{1}(3x)^7(2)^1 = 8 \cdot 2187x^7 \cdot 2 = \mathbf{34992x^7}$ $T_3 = \binom{8}{2}(3x)^6(2)^2 = 28 \cdot 729x^6 \cdot 4 = \mathbf{81648x^6}$ **First three terms:** $6561x^8 + 34992x^7 + 81648x^6$	**Scale 15D (0, 4, 6, 8, 15)** *Low Partial Credit* • Evidence of $\binom{8}{r}$; correct formula *Mid Partial Credit* • One term fully correct *High Partial Credit* • Two terms fully correct *Full Credit* −1 • Ascending powers of x – apply *

Section B Contexts and Applications 150 marks

Q7 Model Solution – 50 Marks		Marking Notes
(a)(i)	Volume: $V = \pi r^2 h = 400\pi \Rightarrow h = \frac{400}{r^2}$ Surface area (base + curved side): $S = \pi r^2 + 2\pi r h = \pi r^2 + 2\pi r \cdot \frac{400}{r^2} = \pi r^2 + \frac{800\pi}{r}$ $S(r) = \pi r^2 + \frac{800\pi}{r}$ ✓	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Volume equation or S with h present *High Partial Credit* • Correct substitution for h

Q7 (cont.)		Marking Notes
(a)(ii)	$S'(r) = 2\pi r - \dfrac{800\pi}{r^2}$	**Scale 5B (0, 2, 5)** *Partial Credit* • Work of merit differentiating $S(r)$ *Full Credit* -1 • π omitted throughout – apply *
(a)(iii)	$S'(r) = 0$: $2\pi r = \dfrac{800\pi}{r^2} \Rightarrow 2r^3 = 800 \Rightarrow r^3 = 400$ $r = \sqrt[3]{400} \approx 7.37$ cm $S''(r) = 2\pi + \dfrac{1600\pi}{r^3}$ $S''(\sqrt[3]{400}) = 2\pi + \dfrac{1600\pi}{400} = 2\pi + 4\pi = 6\pi > 0$ ✓ **minimum**	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Sets $S'(r) = 0$ with merit *Mid Partial Credit* • Solves for $r = \sqrt[3]{400}$ *High Partial Credit* • Correct r; S'' verification begun *Full Credit* -1 • r correct but S'' check missing – apply *
(a)(iv)	$S(\sqrt[3]{400}) = \pi(400)^{2/3} + \dfrac{800\pi}{400^{1/3}}$ $= \pi \cdot 400^{2/3} + 800\pi \cdot 400^{-1/3}$ $= 3\pi \cdot 400^{2/3}$ $\approx 3\pi \cdot (7.368)^2 \approx 3\pi \cdot 54.29 \approx 512$ cm^2 (accept 511–513)	**Scale 5B (0, 2, 5)** *Partial Credit* • Substitutes $r = \sqrt[3]{400}$ into $S(r)$ *Full Credit* -1 • Not rounded to nearest cm^2 – apply *
(b)(i)	$s(t) = 2t^3 - 9t^2 + 12t - 1$ $v(t) = s'(t) = 6t^2 - 18t + 12$ $a(t) = v'(t) = 12t - 18$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • One correct derivative *High Partial Credit* • Both correct with minor error
(b)(ii)	$v(t) = 0$: $6t^2 - 18t + 12 = 0$ $t^2 - 3t + 2 = 0$ $(t-1)(t-2) = 0$ **Particle at rest at** $t = 1$ s **and** $t = 2$ s	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Sets $v(t) = 0$ *High Partial Credit* • Correct quadratic; one root found
(b)(iii)	$s(0) = 0 - 0 + 0 - 1 = -1$ $s(1) = 2 - 9 + 12 - 1 = 4$ $s(2) = 16 - 36 + 24 - 1 = 3$ $s(3) = 54 - 81 + 36 - 1 = 8$ $d = \lvert s(1) - s(0)\rvert + \lvert s(2) - s(1)\rvert + \lvert s(3) - s(2)\rvert$ $= \lvert 4 - (-1)\rvert + \lvert 3 - 4\rvert + \lvert 8 - 3\rvert$ $= 5 + 1 + 5$ **Total distance** $= 11$ m	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Evaluates s at some values of t *High Partial Credit* • Correct s values and direction changes *Full Credit* -1 • $s(3) - s(0) = 9$ only – apply *

Q8 Model Solution – 50 Marks		Marking Notes
(a)(i)	$f(x) = \dfrac{x^2 - 16}{x^2 - 4x} = \dfrac{(x-4)(x+4)}{x(x-4)}$ $f(x) = \dfrac{x+4}{x}, \quad x \neq 0$ and $x \neq 4$ (The value $x = 4$ must be excluded even though it cancels.)	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Factorises numerator or denominator *High Partial Credit* • Correct simplified form; excluded value missing

Q8 (cont.)		Marking Notes
(a)(ii)	$\lim_{x\to\infty} \frac{x+4}{x} = \lim_{x\to\infty}\left(1+\frac{4}{x}\right) = \mathbf{1}$ **Geometric interpretation:** The graph of $y = f(x)$ has a **horizontal asymptote** at $y = 1$; as $x \to \pm\infty$, the curve approaches the line $y = 1$ but never reaches it.	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Evaluates limit with merit *High Partial Credit* • Correct limit; interpretation missing
(a)(iii)	$f(x) = 1 + \frac{4}{x}$ $f'(x) = -\frac{4}{x^2}$ Since $x^2 > 0$ for all $x \neq 0$, we have $f'(x) = -\frac{4}{x^2} < 0$ for all x in the domain. $\therefore$ f is **strictly decreasing** on its entire domain. ✓	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Differentiates simplified form *High Partial Credit* • Correct $f'(x) < 0$; conclusion not stated
(b)(i)	After bounce 1: $4 \times \frac{3}{4} = \mathbf{3}$ m After bounce 2: $3 \times \frac{3}{4} = \frac{\mathbf{9}}{\mathbf{4}}$ m After bounce 3: $\frac{9}{4} \times \frac{3}{4} = \frac{\mathbf{27}}{\mathbf{16}}$ m	**Scale 5B (0, 2, 5)** *Partial Credit* • One height correct
(b)(ii)	$\frac{9/4}{3} = \frac{3}{4}$ and $\frac{27/16}{9/4} = \frac{27}{16} \times \frac{4}{9} = \frac{3}{4}$ Ratio is constant $\Rightarrow$ heights form a **geometric sequence** ✓ **Common ratio** $r = \frac{\mathbf{3}}{\mathbf{4}}$	**Scale 5B (0, 2, 5)** *Partial Credit* • Shows constant ratio *Full Credit* -1 • States r without showing constant – apply *
(b)(iii)	First term of bounce heights $= 3 = 4 \cdot \frac{3}{4}$ **Height after nth bounce** $= 4 \cdot \left(\frac{3}{4}\right)^n$	**Scale 5B (0, 2, 5)** *Partial Credit* • Correct formula attempted
(b)(iv)	Sum of all bounce heights (infinite GP, $\lvert r\rvert < 1$): $S_\infty = \frac{3}{1-\frac{3}{4}} = \frac{3}{\frac{1}{4}} = 12$ m Total distance = initial drop + 2×(sum of bounce heights) $= 4 + 2 \times 12 = 4 + 24 = \mathbf{28}$ m	**Scale 15D (0, 4, 6, 8, 15)** *Low Partial Credit* • States S_∞ formula or identifies GP *Mid Partial Credit* • $S_\infty = 12$ obtained *High Partial Credit* • Multiplies by 2; one component missing

Q9 Model Solution – 50 Marks		Marking Notes
(a)(i)	Monthly interest rate $= \frac{3.6\%}{12} = 0.3\% = \mathbf{0.003}$	**Scale 5A (0, 5)** *Note: No partial credit.*
(a)(ii)	$PV_1 = \frac{1455.96}{(1.003)^1}$ $PV_2 = \frac{1455.96}{(1.003)^2}$ $PV_3 = \frac{1455.96}{(1.003)^3}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • One present value correctly expressed *High Partial Credit* • Two present values correct

Q9 (cont.)		Marking Notes
(a)(iii)	Sum of GP with n terms, first term $a = \frac{1455.96}{1.003}$, ratio $r = \frac{1}{1.003}$: $A = \frac{1455.96}{1.003} \cdot \frac{1-\left(\frac{1}{1.003}\right)^n}{1-\frac{1}{1.003}}$ Multiply numerator and denominator of the fraction by 1.003: $A = \frac{1455.96}{1.003} \cdot \frac{1.003\left[1-(1.003)^{-n}\right]}{1.003-1}$ $A = \mathbf{1455.96 \cdot \frac{1-(1.003)^{-n}}{0.003}}$ ✓	**Scale 15D (0, 4, 6, 8, 15)** *Low Partial Credit* • Identifies GP structure *Mid Partial Credit* • S_n formula correctly applied *High Partial Credit* • Denominator simplified correctly
(b)	Set $A = 320\,000$: $320\,000 = 1455.96 \cdot \frac{1-(1.003)^{-n}}{0.003}$ $1-(1.003)^{-n} = \frac{320000 \times 0.003}{1455.96} = \frac{960}{1455.96} \approx 0.6593$ $(1.003)^{-n} \approx 0.3407$ $-n\ln(1.003) = \ln(0.3407)$ $n = \frac{-\ln(0.3407)}{\ln(1.003)} = \frac{1.0778}{0.002996} \approx \mathbf{360}$ months	**Scale 15D (0, 4, 6, 8, 15)** *Low Partial Credit* • Substitutes $A = 320\,000$ *Mid Partial Credit* • Isolates $(1.003)^{-n}$ *High Partial Credit* • Takes logarithms correctly *Full Credit* -1 • Non-integer or unrounded answer – apply *
(c)	Deposits made *now* and monthly, so deposit 1 grows for 12 months, deposit 2 for 11 months, etc. Total $= 800(1.003)^{12} + 800(1.003)^{11} + \cdots + 800(1.003)^1$ $= 800(1.003) \cdot \frac{(1.003)^{12}-1}{1.003-1}$ $= 800 \times 1.003 \times \frac{(1.003)^{12}-1}{0.003}$ $(1.003)^{12} \approx 1.03660$ $= 800 \times 1.003 \times \frac{0.03660}{0.003} = 800 \times 1.003 \times 12.20 \approx \mathbf{9891}$ ✓	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Identifies GP or writes first term *Mid Partial Credit* • First term and ratio both correct *High Partial Credit* • S_n correctly applied; arithmetic error only

Q10	Model Solution – 50 Marks	Marking Notes
(a)	**Prove:** $1^2 + 2^2 + \cdots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$ **Base case** $n = 1$: LHS $= 1^2 = 1$ RHS $= \dfrac{1 \cdot 2 \cdot 3}{6} = 1$ LHS = RHS ✓ **Inductive step:** Assume true for $n = k$: $1^2 + 2^2 + \cdots + k^2 = \dfrac{k(k+1)(2k+1)}{6}$ For $n = k+1$, add $(k+1)^2$ to both sides: $1^2 + \cdots + k^2 + (k+1)^2 = \dfrac{k(k+1)(2k+1)}{6} + (k+1)^2$ $= \dfrac{k(k+1)(2k+1) + 6(k+1)^2}{6}$ $= \dfrac{(k+1)\left[k(2k+1) + 6(k+1)\right]}{6}$ $= \dfrac{(k+1)(2k^2+7k+6)}{6}$ $= \dfrac{(k+1)(k+2)(2k+3)}{6}$ $= \dfrac{(k+1)\left[(k+1)+1\right]\left[2(k+1)+1\right]}{6}$ ✓ True for $n = k+1$. By induction, true for all $n \in \mathbb{N}$. **QED**	**Scale 15D (0, 4, 6, 8, 15)** *Low Partial Credit* • Correct base case *Mid Partial Credit* • Inductive step set up; $(k+1)^2$ added *High Partial Credit* • Simplification begun; factorisation nearly complete *Full Credit* -1 • Correct but conclusion not stated – apply *
(b)(i)	$V = \dfrac{4}{3}\pi r^3$ $\dfrac{dV}{dr} = 4\pi r^2$	**Scale 5B (0, 2, 5)** *Partial Credit* • Attempts differentiation with merit
(b)(ii)	$\dfrac{dV}{dt} = 50 \text{ cm}^3\text{s}^{-1}$ By chain rule: $\dfrac{dV}{dt} = \dfrac{dV}{dr} \cdot \dfrac{dr}{dt}$ $\dfrac{dr}{dt} = \dfrac{dV/dt}{dV/dr} = \dfrac{50}{4\pi r^2}$ At $r = 5$: $\dfrac{dr}{dt} = \dfrac{50}{4\pi \cdot 25} = \dfrac{50}{100\pi} = \dfrac{1}{2\pi}$ cm s^{-1}	**Scale 15D (0, 4, 6, 8, 15)** *Low Partial Credit* • States chain rule *Mid Partial Credit* • Correct expression for $\frac{dr}{dt}$ *High Partial Credit* • Correct expression; substitution error
(b)(iii)	$A = 4\pi r^2 \Rightarrow \dfrac{dA}{dr} = 8\pi r$ $\dfrac{dA}{dt} = \dfrac{dA}{dr} \cdot \dfrac{dr}{dt} = 8\pi r \cdot \dfrac{1}{2\pi} = 4r$ At $r = 5$: $\dfrac{dA}{dt} = 4 \times 5 = 20$ cm^{2}s^{-1}	**Scale 15D (0, 4, 6, 8, 15)** *Low Partial Credit* • Differentiates $A = 4\pi r^2$ *Mid Partial Credit* • Chain rule applied correctly *High Partial Credit* • Correct expression; substitution error